OPEN-SOURCE EDGE AI ENGINEERING STUDY • GITHUB.COM/AJAYAKUMARV/QWEN3STUDIO-LOCALAI

30.5B Sparse MoE (Qwen3-30B-A3B) at 39.2 tok/s on 4 GB VRAM

Intel Core i7-12700K (32 GB RAM) + NVIDIA T1000 (4 GB VRAM) • 100% Offline • <1ms Symbolic+AST Critic

5.7x SPEEDUP
39.2 tok/s
Up from 6.89 tok/s cold baseline

147x FASTER TTFT
291 ms
Down from 42,811 ms mmap stall

0.6B DRAFT BOOST
+45.1%
94.8% accept in Full 8-Expert mode

SYMBOLIC+AST GATE
0.40 ms
100% hallucination trap recovery

System / Configuration (4 GB VRAM)	Decode Speed	Max 4GB Context	Fact & Code Verification
Qwen3Studio (Top-4 + -ncmoe 41 + P-Cores)	**39.2 tok/s (Optimal)**	**8,192 tok (3.76 GB)**	**0.40ms Symbolic+AST + Critic**
Qwen3Studio (All-8 Experts + 0.6B Speculative)	**27.1 tok/s (+45.1%)**	**8,192 tok (3.89 GB)**	**94.8% Draft Token Acceptance**
MIT & Berkeley FreeToken (on 4 GB RTX 2050)	21.3 tok/s (-46%)	4,096 tok (8K Fails)	None (Unverified LLM output)
Dynamic PCIe LRU Cache (--moe-cache-mib 768)	8.57 tok/s (PCIe Stall)	8,192 tok	PCIe cache-miss thrashing
Default Cold Baseline (-ngl 99 --cpu-moe mmap)	6.89 tok/s (42.8s TTFT)	4,096 tok / slot	OS page-fault stalls

Executive Takeaway & Repository: https://github.com/AjayakumarV/Qwen3Studio-LocalAI (Apache-2.0 License)

1. Selective Top-4 Routing + Static 7-Layer GPU Pinning (-ncmoe 41): Achieves 39.2 tok/s and 291 ms TTFT on a 4 GB NVIDIA T1000 + 32 GB RAM (85% faster than MIT FreeToken on 4 GB VRAM).
2. Regime-Aware 0.6B Speculative Decoding: Achieves 94.8% draft token acceptance and +45.1% speedup (27.11 tok/s) in Full 8-Expert mode.
3. Sub-Millisecond (0.403 ms avg) Symbolic & AST Critic (verifier.py): Catches 100% of numerical and syntax hallucinations with zero code execution risk.
4. 100% Air-Gapped Desktop Suite (Qwen3Studio.exe): Includes 0 MB VRAM CPU-INT8 Faster-Whisper STT, offline TTS, and <15 ms BM25 Top-5 multi-format RAG.